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**CHRYSODICTYON INDICUM GEN. ET SP. NOV.,
A NEW MEMBER OF THE CHRYSOPHYCEAE
FROM SOUTH INDIA**

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CHRYSODICTYON INDICUM GEN. ET SP. NOV., A NEW MEMBER OF THE CHRYSOPHYCEAE FROM SOUTH INDIA

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(With one Plate and eight Figs. in the Text)

AN account is given in this note of an alga belonging to the Chrysophyceae which was found growing at Madras. Records of Chrysophyceae from India are very few. In fact there appears to be so far only one record from India, viz., two forms of *Hydrurus foetidus* (Vill.) Kirchn. from Kashmir. Bharadwaja (1936) found these two forms in a shallow fast flowing stream formed by the melting of snow near Sonamarg, about 9,000 ft. above sea-level.

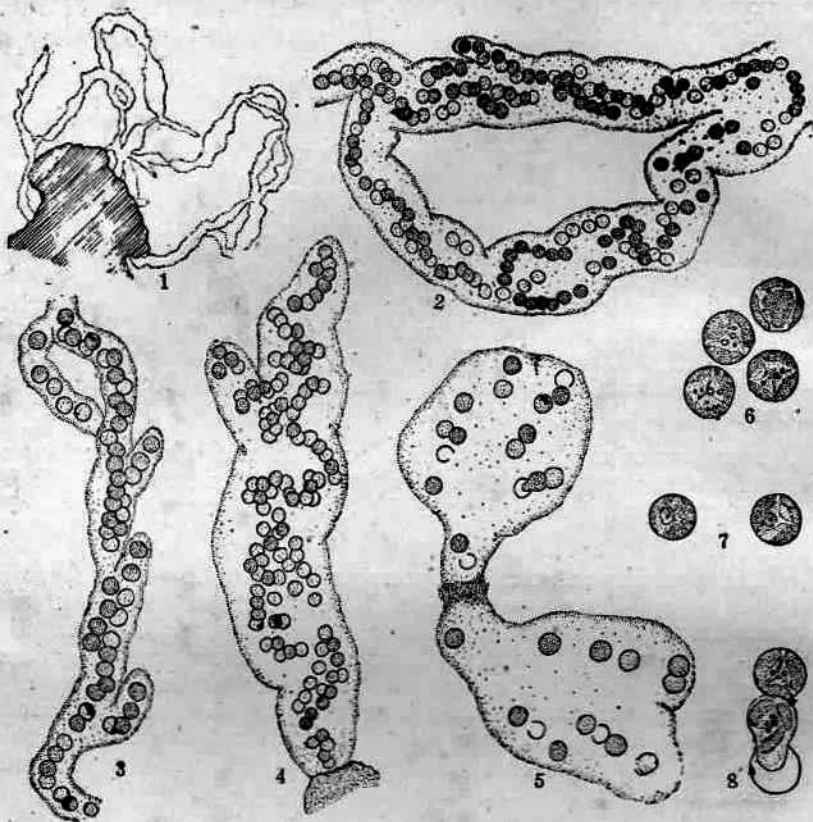
Members of this group generally occur in cold winter weather or in cold streams and pools of mountainous tracts (Fritsch, 1935, p. 507). Its occurrence in a tropical area like Madras is, therefore, extremely interesting.

The alga was collected in December 1939 at Adyar, a suburb of Madras, where it was growing attached to some debris along with other algae in a thin stream of very clear water trickling out of a paddy-field into an adjoining drain. This portion was shaded by some trees growing near it. The alga was collected again in January 1942 in the same locality under more or less similar environmental conditions. The colder part of the year (December and January) when it occurred and the shady situation in which it was found are evidently ideal conditions for its occurrence in a tropical area.

The thallus of the alga when fully developed consists of a thin, transparent, more or less cylindrical mucilaginous net-work, enclosing a large number of small, rounded cells, which under low power, appear as a golden brown chain-like net-work inside the mucilaginous matter (Pl. I, Fig. 1). The thallus is somewhat branched, the branches varying from short lateral lobes to long loops inter-connecting various portions of the thallus (Pl. I, Figs. 1, 2; Text-figs. 1-3). In young and undeveloped colonies, the thallus is somewhat thin and cylindrical with a single row of cells centrally placed (Pl. I, Fig. 5) or is somewhat sac-like and feebly branched containing a number of cells peripherally distributed (Pl. I, Fig. 6; Text-figs. 4-5). In all cases the thallus is attached at one end by the general mucilaginous substance to any piece of debris or other algae occurring along with it (Pl. I, Figs. 1, 5, 6; Text-figs. 1, 4).

The arrangement of cells inside the mucilaginous matrix is somewhat characteristic and gives a distinctive appearance to the alga.

In portions of the thallus, where the mucilage is thin and narrow, the cells are arranged centrally in a single series, one behind the other (Pl. I, Fig. 5; Text-fig. 3), while in portions where the mucilage is broad



Text-Figs. 1-8. *Chrysodictyon indicum* gen. et sp. nov. Fig. 1. A fully developed colony, showing the net-like habit (semi-diagrammatic). Fig. 2. Portion of the thallus showing the arrangement of cells inside the mucilaginous matrix. Note the serial arrangement of cells often in groups of 2, 4 or 8. Fig. 3. Another portion of the mature thallus showing the development of branches and loops. Note again the tendency of cells to be grouped in multiples of two and four. Figs. 4 & 5. Two young thalli in early stages of development. Fig. 4. Illustrates a young thallus attached at one end and showing incipient branching at the other. Note the grouping of cells particularly near the upper end. Fig. 5. Shows a young colony attached in the middle and containing very few cells, the latter often grouped in pairs with a ruptured, empty wall lying close by. In one case the protoplast is still in the process of escaping out of the mother wall (cf. text). Fig. 6. Four individual cells showing 3-4 plate-like, parietal chromatophores and refractive, leucosin (?) granules. Fig. 7. Two cells after staining with iodine, showing the chromatophores and the nucleus. Fig. 8. A probable case of germinating cyst, showing one of the daughter protoplasts already rounded and the other just in the process of escaping out of the ruptured wall. (All figures drawn from living material except Fig. 7. Fig. 1. $\times 24$; Fig. 2. $\times 192$; Figs. 3-5, $\times 264$; Figs. 6-8, $\times 856$.)

and sac-like, they are distributed somewhat peripherally, in small serial groups of 2, 4 or 8, often running parallel to one another or tending to be connected together into a sort of a coarse reticulum (Pl. I, Figs. 1-3; Text-figs. 2, 3, 4). However, in certain portions of the thallus they appear irregularly distributed, evidently due to displacement by copious secretion of mucilage between them.

The individual cells of the alga are quite spherical and contain each 2-3 or at the most 4, parietal, plate-like chromatophores, golden brown in colour and a single nucleus (Pl. I, Figs. 2-4; Text-figs. 6-8). A number of small refractive granules, probably consisting of leucosin, are seen in the hyaline cytoplasm. Neither a contractile vacuole nor an eye-spot is present. The cells range in size from 6.7μ to 11.7μ in diameter, their size varying evidently with the age and portion of the thallus. They are generally larger in well grown colonies and in less crowded portions, where they measure $8.35-11.7\mu$ in diameter, while in younger colonies and in crowded portions they are somewhat smaller and measure $6.7\mu-10.02\mu$ in diameter. Probably some of the larger cells are those about to divide, while the smaller ones are products of recent division.

The growth of the thallus takes place evidently by the repeated division of all the cells constituting it. While no actual division has been observed, the frequent association of cells in groups of two, four, eight and so on, arranged either serially or in parallel rows in different portions of the thallus (Pl. I, Figs. 1-3, 5; Text-figs 2-5) suggests the probability of their derivation through the successive division of all the cells, first into two, then into four and so on. As a matter of fact, a careful examination of the entire thallus appears to indicate that it is evidently composed of a number of such serial groups merged together by their mucilaginous investments into a single thalloid net-work (Pl. I, Figs. 1, 2). Staining with methylene blue or mucilage stains, however, fails to show any striations or the limits of the individual mucilaginous investment around these several groups of cells. Probably they are so thin that they are completely merged in the general mucilaginous investment.

Neither cysts nor motile reproductive bodies have been observed, even though the alga was kept under observation for some time. However, in a young thallus, containing a small number of cells, a few empty ruptured walls were seen lying here and there, close to pairs of cells (Text-fig. 5), suggesting a derivation of these cells, either from germinating cysts or from ordinary cells, whose protoplast after division into two appeared to have escaped out, discarding their mother wall. In one of these cases, one of the two protoplasts was seen still in the process of rounding up just outside the ruptured mother wall (Pl. I, Fig. 4; Text-figs. 5, 8), while the other had already rounded up.

The present alga in the reticulate nature of its thallus, in the arrangement of the cells in groups of 2, 4, 8 and so on and in the presence of 2-3 or even 4 plate-like parietal chromatophores in each cell, differs from any other member of the palmelloid Chrysophyceae recorded so far. In the general appearance of the thallus, it somewhat resembles

Celloniella, a gelatinous member of the Chrysocapsaceæ recorded by Pascher (1929) from certain fast flowing streams in Central Europe. But it differs from *Celloniella* in the reticulate nature of its thallus, in the absence of a distinct stalk composed of firmer mucilage and also an attaching disc which is markedly stratified and encrusted with carbonate of lime, in the growth of the thallus by the division of each and every cell and finally in the presence of 2-3 or even 4 chromatophores in each cell. Further no cysts or motile swimmers similar to those recorded in *Celloniella* have been observed in the present alga. The differences are many and wide, and it is therefore not possible to include the present alga in *Celloniella*, in spite of the fact that the latter form has been shown by Pascher (1929) to assume various shapes according to the conditions under which it grows. While it is possible that some of the differences might have been brought about by the marked differences in the environmental conditions under which the two algae have been found, still the remaining differences are so fundamental that it is advisable to keep the present alga in a new genus, quite distinct from *Celloniella*, though somewhat close to it. The alga may be named *Chrysodictyon indicum* gen. et sp. nov.

DESCRIPTION

Chrysodictyon gen. nov.

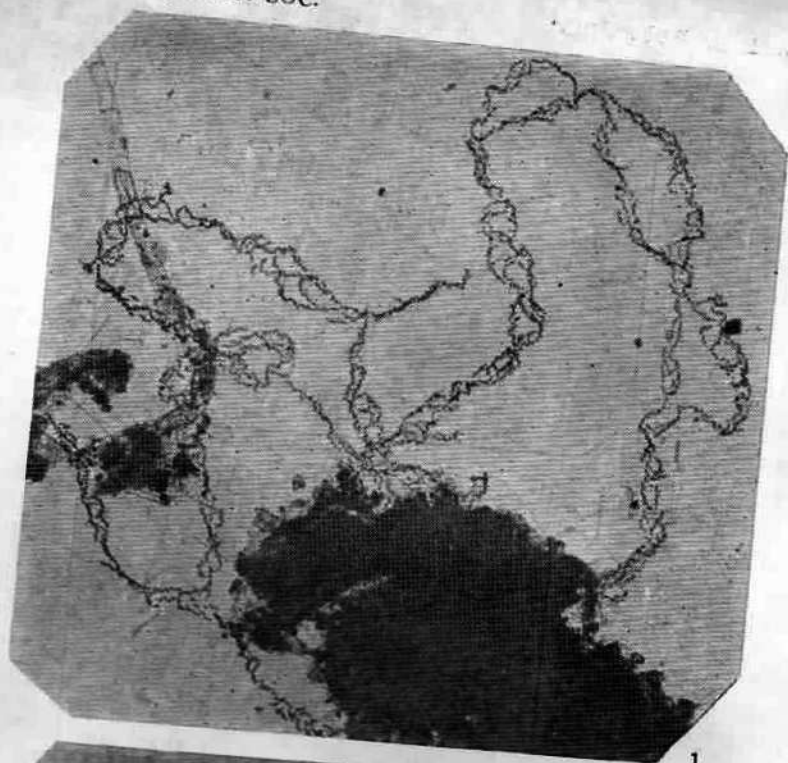
Thallus mucilaginous, net-like when fully developed, thin, cylindrical or sac-like when young: thallus somewhat branched, branches varying from short lateral lobes to long loops interconnecting various portions of the thallus: thallus attached at one end by the general mucilaginous substance: cells numerous and embedded in the gelatinous matrix, often in series or groups of two, four, eight and so on, the groups of cells often arranged to form a coarse net-work: cells spherical, each containing 2-3 or even 4 plate-like, parietal, golden brown chromatophores and a single nucleus: growth of the thallus by the successive division of each and every cell: reproduction not known.

Chrysodictyon indicum sp. nov.

General characters same as those of the genus. Thallus when fully developed up to 0.5 cm. in length: cells 6.7-11.7 μ in diameter: reproduction unknown.

Habitat.—Attached to some debris along with other algae in a thin stream of water trickling out of a paddy field into an adjoining drain at Adyar, a suburb of Madras.

In conclusion, I wish to express my sincere gratitude to Professor M. O. P. Iyengar, M.A., Ph.D. (Lond.), F.L.S., for his kind help and guidance during the course of this investigation.

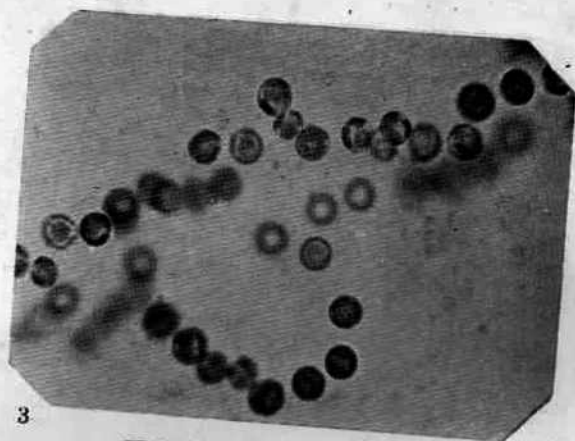


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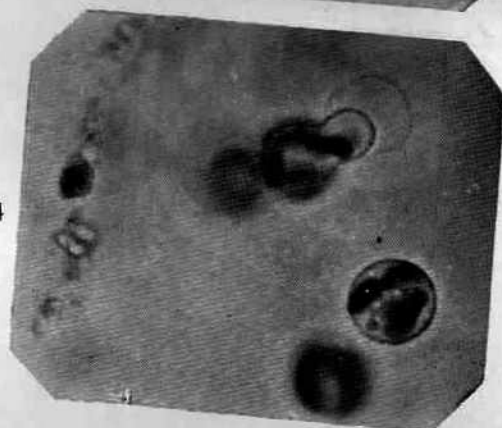


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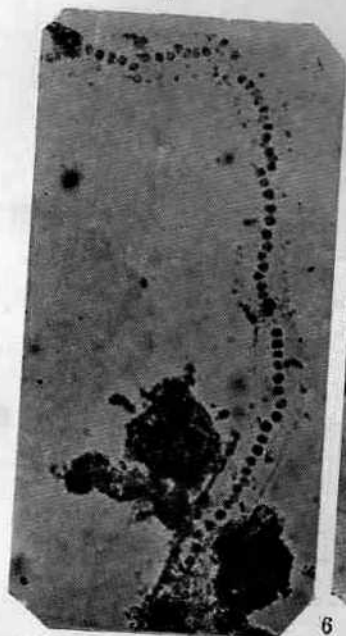


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EXPLANATION OF PLATES

Chrysodictyon indicum gen. et sp. nov.

Fig. 1. Photomicrograph of a well grown thallus. $\times 60$.

Fig. 2. A portion of the thallus showing the general arrangement of the cells inside the mucilaginous matrix. Note the serial arrangement of cells in groups of 2, 4, or 8 and their tendency to form a reticulate pattern. $\times 220$.

Fig. 3. A small portion of the thallus further enlarged, to show the structure of individual cells. Note the presence of 3-4 parietal, plate-like chromatophores inside each cell and also the tendency of the cells to be grouped in fours. $\times 450$.

Fig. 4. A probable case of a germinating cyst showing one of the daughter-protoplasts still in the process of escaping out, with the ruptured mother-wall lying nearby. Note the three distinct chromatophores and the refractive granules in the cell close by. $\times 1200$.

Fig. 5. A young thallus attached at one end to the debris and showing the narrow, cylindrical mucilaginous matrix enclosing a single row of cells. $\times 140$.

Fig. 6. Another young thallus, sac-like showing the distribution of cells peripherally in groups. $\times 300$.

(All figures from living material.)